

Antichains among Divisor Sums of Divisors

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Abstract

For a positive integer n , let

$$S(n) = \{\sigma(d) : d \mid n\},$$

where σ is the sum-of-divisors function, and let $a(n)$ be the width of $S(n)$ under divisibility. We determine $a(n)$ for prime powers and, for every fixed exponent pattern, determine the largest possible value of $a(n)$. The maximum is the width of a finite product of divisibility posets and is attained for infinitely many choices of the underlying primes. In the squarefree case we go further: if $\omega(n) = r$, then every integer from 1 to $\binom{r}{\lfloor r/2 \rfloor}$ occurs as $a(n)$, and every such value occurs infinitely often.

For equal exponent patterns the problem leads to powers of the divisibility poset

$$D_m = (\{1, \dots, m\}, |).$$

Using Engel's theory of optimal representations, we determine the representation variance of D_m . The main structural step is an exact reduction to the prime directions 2 and 3. We show that these two directions are both sufficient and necessary in a universal sense, and we identify the underlying Ferrers threshold. This yields an exact formula for $V(D_m)$, the limit

$$V(D_m) \longrightarrow 1.036793420159825251 \dots,$$

and an explicit asymptotic constant for the maximal width in equal-exponent families.

1 Introduction

For a positive integer n , consider

$$S(n) = \{\sigma(d) : d \mid n\}.$$

We order this finite set by divisibility and write

$$a(n) = \text{width}(S(n), |).$$

Thus $a(n)$ is the largest number of distinct divisor sums, taken from divisors of the same integer n , such that no one divides another.

For example, if $n = 6$,

$$S(6) = \{1, 3, 4, 12\}.$$

The two middle values are incomparable, so $a(6) = 2$. The initial values of the sequence are

$$1, 1, 1, 2, 1, 2, 1, 2, 2, 1, 1, 3, 1, 2, 2, 3, 1, 3, 1, 2, \dots$$

Sets with no internal divisibility are usually called *primitive sets*; they form a classical meeting point of multiplicative number theory and extremal combinatorics [1, 2]. Our setting is more

constrained. We do not choose integers freely: we first take all divisors of one integer, apply σ , and then study the divisibility relations among the resulting values. The arithmetic map

$$d \mapsto \sigma(d)$$

therefore sends the divisor lattice of n into the divisibility poset of the positive integers, and the basic question is how much order structure survives.

For one prime factor the answer is simple. Since

$$\sigma(p^{k-1}) = 1 + p + \cdots + p^{k-1},$$

divisibility is controlled exactly by divisibility of the index k . With several prime factors, however, products coming from different prime bases can divide one another. These extra *cross-divisibilities* are the main arithmetic difficulty in the paper.

The first part asks how far they can be avoided. For a fixed exponent pattern

$$\mathbf{e} = (e_1, \dots, e_r),$$

we show that the primes can be chosen so that no divisibility relation survives except those forced by the indices. The maximal problem therefore becomes a purely combinatorial width problem for the universal product poset

$$D_{e_1+1} \times \cdots \times D_{e_r+1}.$$

This is Theorem 4.1.

The squarefree case then raises a finer question. The maximum is the central binomial coefficient and the minimum is 1, but must every value in between occur? We prove that the answer is yes. The proof does not merely interpolate numerically: it constructs one new prime factor whose cross-divisibilities with the old squarefree structure are governed by a single weighted threshold. Moving that threshold one step at a time changes the width by at most one, which forces the entire interval of values.

The second half of the paper starts from equal exponent patterns. They lead to large powers of

$$D_m = (\{1, \dots, m\}, |),$$

so the problem enters Engel's theory of optimal representations and limit Sperner theorems [5, 8]. Divisibility orders on initial intervals have also been studied through other invariants, in particular order dimension [9, 10]. Here the relevant invariant is the minimum representation variance $V(D_m)$.

The full divisibility order is not needed to determine this variance. The directions 2 and 3 already force the optimum. We prove that this reduction is exact for every m , that no smaller universal prime core works, and that the special role of 2 and 3 comes from a sharp Ferrers-slope threshold. This leads to an exact finite formula for $V(D_m)$ and to its limiting constant.

These three parts are linked by the role of cross-divisibilities. In the maximal problem they are eliminated, in the squarefree spectrum they are controlled, and in the variance problem the active directions are classified. The minimum-width problem, where cross-divisibilities must instead be exploited as strongly as possible, is left open.

2 Basic properties and prime powers

We begin with an elementary monotonicity property.

Proposition 2.1. *If $m \mid n$, then*

$$a(m) \leq a(n).$$

Proof. Every divisor of m is a divisor of n , hence $S(m) \subseteq S(n)$. Every antichain in $S(m)$ is therefore an antichain in $S(n)$. $\square$

For a prime p , put

$$R_k(p) = 1 + p + \cdots + p^{k-1} = \frac{p^k - 1}{p - 1}.$$

Lemma 2.2. *For $i, j \geq 1$,*

$$R_i(p) \mid R_j(p) \iff i \mid j.$$

Proof. If $i \mid j$, the divisibility follows from the factorization of $x^j - 1$. Conversely,

$$\gcd(R_i(p), R_j(p)) = R_{\gcd(i, j)}(p).$$

Thus $R_i(p) \mid R_j(p)$ forces $i = \gcd(i, j)$, hence $i \mid j$. $\square$

Theorem 2.3. *For every prime p and every $e \geq 0$,*

$$a(p^e) = \left\lceil \frac{e+1}{2} \right\rceil.$$

Proof. Since

$$S(p^e) = \{R_1(p), \dots, R_{e+1}(p)\},$$

Lemma 2.2 identifies this poset with D_{e+1} . The integers

$$\left\lfloor \frac{e+1}{2} \right\rfloor + 1, \dots, e+1$$

form an antichain. Conversely, partition $1, \dots, e+1$ according to odd part; every class is a chain of the form $u, 2u, 4u, \dots$. There are $\lceil (e+1)/2 \rceil$ such chains, so Dilworth's theorem gives the result. $\square$

Corollary 2.4. *Every positive integer occurs as a value of $a(n)$.*

Proof. For $k \geq 1$, Theorem 2.3 gives $a(p^{2k-2}) = k$. $\square$

Example 2.1 (Two small divisibility posets). For $n = p^4$, the five values $R_1(p), \dots, R_5(p)$ have exactly the divisibility relations of D_5 . In particular,

$$\{R_3(p), R_4(p), R_5(p)\}$$

is a largest antichain. For $n = pq$, the four values form either a chain or a diamond, according as $p+1$ divides $q+1$ or not. Figure 1 displays these two basic shapes.

Proposition 2.5. *If $a(n) = 1$, then n is squarefree.*

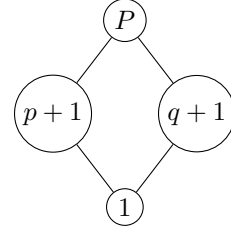
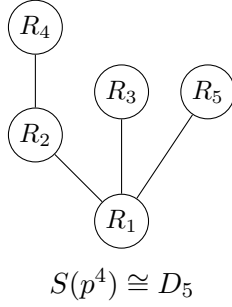


Figure 1: Two elementary examples. On the left, the prime-power case is the divisibility poset on the indices. On the right, the squarefree two-prime case is a diamond when the two middle values are incomparable.

Proof. If $p^2 \mid n$, then $p+1$ and p^2+p+1 belong to $S(n)$. The latter is larger than $p+1$ and is congruent to 1 modulo $p+1$, so these two values are incomparable. $\square$

Proposition 2.6. *Let $p < q$ be primes. Then*

$$a(pq) = \begin{cases} 1, & p+1 \mid q+1, \\ 2, & p+1 \nmid q+1. \end{cases}$$

Proof. We have

$$S(pq) = \{1, p+1, q+1, (p+1)(q+1)\}.$$

The only possible incomparability is between $p+1$ and $q+1$. $\square$

3 Squarefree integers

Suppose

$$n = p_1 \cdots p_r$$

is squarefree. For $I \subseteq [r]$,

$$\sigma\left(\prod_{i \in I} p_i\right) = \prod_{i \in I} (p_i + 1).$$

Theorem 3.1. *If n is squarefree and $\omega(n) = r$, then*

$$a(n) \leq \binom{r}{\lfloor r/2 \rfloor}.$$

Proof. Choose one subset $I \subseteq [r]$ representing each value in an antichain of $S(n)$. If $I \subseteq J$, then the product indexed by I divides the product indexed by J , which is impossible for two distinct values in the chosen antichain. The chosen subsets therefore form an antichain in the Boolean lattice B_r . Sperner's theorem gives the bound. $\square$

Theorem 3.2. *For every $r \geq 1$, infinitely many squarefree integers n satisfy*

$$\omega(n) = r, \quad a(n) = 1.$$

Proof. The assertion is clear for $r = 1$. Proceed by induction. Suppose n is squarefree and $S(n)$ is a chain, and put $M = \sigma(n)$. By Dirichlet's theorem choose a prime $q \nmid n$ such that

$$q \equiv -1 \pmod{M}.$$

Then

$$S(nq) = S(n) \cup (q+1)S(n).$$

The two copies are chains, and every element of the first copy divides every element of the second because every element of $S(n)$ divides M and hence divides $q+1$. Thus the union is again a chain. $\square$

4 Fixed exponent patterns

The prime-power case suggests a natural model for a general exponent vector. If

$$n = p_1^{e_1} \cdots p_r^{e_r},$$

then every divisor sum is a product of repunits $R_k(p_i)$. If one ignores divisibilities between different prime bases, the index set is the product poset

$$P_{\mathbf{e}} = D_{e_1+1} \times \cdots \times D_{e_r+1}, \quad \mathbf{e} = (e_1, \dots, e_r).$$

The next theorem shows that this naive model is the exact maximal one: all additional cross-divisibilities can be avoided simultaneously.

Theorem 4.1. *For every fixed exponent vector $\mathbf{e}$,*

$$\boxed{\max_{p_1, \dots, p_r} a(p_1^{e_1} \cdots p_r^{e_r}) = \text{width}(P_{\mathbf{e}}),}$$

where the maximum is over distinct primes. Moreover, the maximum is attained for infinitely many choices of $p_1, \dots, p_r$.

Proof. For

$$\mathbf{k} = (k_1, \dots, k_r) \in P_{\mathbf{e}}$$

define

$$F(\mathbf{k}) = \prod_{i=1}^r R_{k_i}(p_i).$$

If $k_i \mid \ell_i$ for every i , then $F(\mathbf{k}) \mid F(\boldsymbol{\ell})$. Given an antichain among the distinct values $F(\mathbf{k})$, choose one index vector representing each value. No two chosen vectors can be comparable in $P_{\mathbf{e}}$, so they form an antichain. Therefore

$$a(p_1^{e_1} \cdots p_r^{e_r}) \leq \text{width}(P_{\mathbf{e}}).$$

It remains to realize exactly the coordinatewise divisibility relations. Put

$$m = \max_i (e_i + 1).$$

For every pair (i, k) with $2 \leq k \leq e_i + 1$, choose a distinct prime $q_{i,k} > m$ satisfying

$$q_{i,k} \equiv 1 \pmod{k},$$

and choose $g_{i,k}$ of exact multiplicative order k modulo $q_{i,k}$. For each fixed i , prescribe

$$p_i \equiv g_{i,k} \pmod{q_{i,k}}$$

for the primes belonging to the i -th coordinate, and

$$p_i \equiv 1 \pmod{q_{j,k}} \quad (j \neq i)$$

for all primes belonging to the other coordinates. The Chinese remainder theorem gives a reduced residue class, and Dirichlet's theorem supplies infinitely many primes in it. The p_i may therefore be chosen distinct.

For $2 \leq k \leq e_i + 1$ and $1 \leq u \leq e_i + 1$,

$$q_{i,k} \mid R_u(p_i) \iff k \mid u,$$

because p_i has exact order k modulo $q_{i,k}$. On the other hand, if $j \neq i$, then

$$R_u(p_j) \equiv u \pmod{q_{i,k}},$$

and $1 \leq u \leq m < q_{i,k}$, so $q_{i,k}$ divides none of the relevant factors from another coordinate.

Thus $q_{i,k}$ is a private witness for divisibility by k in the i -th coordinate. If some k_i does not divide ℓ_i , then q_{i,k_i} divides $F(\mathbf{k})$ but not $F(\ell)$. Consequently

$$F(\mathbf{k}) \mid F(\ell) \iff k_i \mid \ell_i \text{ for all } i.$$

Hence the divisibility poset is isomorphic to $P_{\mathbf{e}}$, proving equality. Since each prescribed residue class contains infinitely many primes, infinitely many choices attain the maximum. $\square$

Thus, for a fixed exponent pattern, maximizing $a(n)$ is completely reduced to the combinatorics of the universal poset $P_{\mathbf{e}}$. The arithmetic input is used to realize that poset exactly.

Example 4.1 (A concrete p^2q^2 realization). For the exponent pattern $(2, 2)$ the universal poset is D_3^2 , whose rank numbers are 1, 4, 4 and whose width is 4. The private-prime construction can be made completely explicit. Take

$$q_{1,2} = 5, \quad q_{1,3} = 7, \quad q_{2,2} = 11, \quad q_{2,3} = 13.$$

The primes

$$p = 3719, \quad q = 11131$$

satisfy

$$p \equiv (4, 2, 1, 1) \pmod{(5, 7, 11, 13)}, \quad q \equiv (1, 1, 10, 3) \pmod{(5, 7, 11, 13)}.$$

Here 4 has order 2 modulo 5, 2 has order 3 modulo 7, 10 has order 2 modulo 11, and 3 has order 3 modulo 13. Thus the four auxiliary primes are private witnesses for the two coordinates exactly as in the proof above. Consequently

$$a(3719^2 11131^2) = \text{width}(D_3^2) = 4.$$

This example illustrates the role of the construction: it does not merely produce a large antichain, but removes every divisibility relation not already forced by the exponent indices.

Corollary 4.2. *For every $r \geq 1$,*

$$\boxed{\max_{\substack{n \text{ squarefree} \\ \omega(n)=r}} a(n) = \binom{r}{\lfloor r/2 \rfloor}.}$$

The maximum is attained infinitely often.

Proof. For the squarefree pattern, $P_e = D_2^r$, which is the Boolean lattice B_r . □

Combining Theorem 3.2 and Corollary 4.2 shows that both squarefree extremal values are sharp and occur infinitely often.

5 The complete squarefree spectrum

The preceding results determine the two ends of the squarefree spectrum. What remains is an interpolation question: can a width be skipped?

For

$$\mathcal{W}_r = \{a(n) : n \text{ squarefree}, \omega(n) = r\}$$

and

$$b_r = \binom{r}{\lfloor r/2 \rfloor},$$

the result is

$$\mathcal{W}_r = \{1, 2, \dots, b_r\}.$$

The proof has two ingredients. First, a one-prime extension can preserve a given width. Second, a weighted construction realizes every new width between b_r and b_{r+1} .

Lemma 5.1 (Controlled one-prime extension). *Let n be squarefree, put $N = \sigma(n)$, and let $M \mid N$. Assume that $2 \mid M$ whenever $2 \mid N$. Then there are infinitely many primes $q > N$ for which*

$$q + 1 = Mk, \quad \gcd(k, N) = 1.$$

For every such q and all $x, z \in S(n)$,

$$\boxed{x \mid (q+1)z \iff \frac{x}{\gcd(x, z)} \mid M.} \tag{5.1}$$

Proof. We first choose a residue class c modulo N . For every prime $\ell \mid N$ with $\ell \mid M$, require $c \not\equiv 0 \pmod{\ell}$. For every prime $\ell \mid N$ with $\ell \nmid M$, the prime ℓ is odd by the hypothesis on the factor 2; choose c modulo ℓ avoiding both 0 and M^{-1} . These local conditions can be lifted to the prime-power factors of N , and the Chinese remainder theorem gives c such that

$$\gcd(c, N) = 1, \quad \gcd(Mc - 1, N) = 1.$$

Since also $Mc - 1 \equiv -1 \pmod{M}$, we have

$$\gcd(Mc - 1, MN) = 1.$$

Dirichlet's theorem therefore gives infinitely many primes

$$q \equiv Mc - 1 \pmod{MN}.$$

Choosing one with $q > N$, write

$$q + 1 = M(c + Nt) = Mk.$$

Then $\gcd(k, N) = 1$.

Because n is squarefree, every element of $S(n)$ divides N . Put

$$g = \gcd(x, z), \quad x = gx', \quad z = gz',$$

so that $\gcd(x', z') = 1$. Then

$$x \mid (q + 1)z \iff x' \mid Mkz'.$$

Now $x' \mid N$, hence $\gcd(x', k) = 1$, and $\gcd(x', z') = 1$. Thus the last divisibility is equivalent to $x' \mid M$, which is exactly (5.1). $\square$

Corollary 5.2. *For every $r \geq 1$,*

$$\mathcal{W}_r \subseteq \mathcal{W}_{r+1}.$$

More precisely, if a width occurs for a squarefree integer with r prime factors, then it occurs for infinitely many squarefree integers with $r + 1$ prime factors.

Proof. Apply Lemma 5.1 with $M = N$. Then every $x, z \in S(n)$ satisfy

$$x \mid (q + 1)z,$$

so

$$S(nq) = S(n) \oplus (q + 1)S(n)$$

is the ordinal sum of two copies of $S(n)$. Its width is therefore $a(n)$. The lemma gives infinitely many choices of q . $\square$

We next construct the missing upper part of the spectrum. The idea is to start from an exact Boolean lattice and then add one more prime factor. The new prime creates cross-relations between two Boolean copies, but those relations will be controlled by one integer threshold.

Fix $r \geq 1$ and put

$$A = 2^r, \quad w_i = A + 2^{i-1} \quad (1 \leq i \leq r).$$

Choose distinct auxiliary primes $\ell_1, \dots, \ell_r > 3$. By the Chinese remainder theorem and Dirichlet's theorem, we may choose distinct primes $p_1, \dots, p_r > 3$ such that

$$p_i \equiv 3^{w_i} - 1 \pmod{3^{w_i+1}}, \tag{5.2}$$

$$p_i \equiv -1 \pmod{\ell_i}, \quad p_i \equiv 1 \pmod{\ell_j} \quad (j \neq i). \tag{5.3}$$

The residue classes are reduced, so infinitely many choices are available. Put

$$n = p_1 \cdots p_r, \quad x_I = \prod_{i \in I} (p_i + 1) \quad (I \subseteq [r]).$$

Equation (5.3) makes ℓ_i a private prime divisor of $p_i + 1$. Consequently

$$x_I \mid x_J \iff I \subseteq J, \tag{5.4}$$

so $S(n)$ is exactly a Boolean lattice B_r . By (5.2),

$$v_3(p_i + 1) = w_i.$$

Define

$$W(I) = v_3(x_I) = A|I| + \sum_{i \in I} 2^{i-1}. \quad (5.5)$$

These weights are pairwise distinct integers. Indeed, for fixed $|I|$ the binary sum $\sum_{i \in I} 2^{i-1}$ determines I , while different cardinalities lie in disjoint intervals $[kA, (k+1)A - 1]$. Let

$$H = W([r]) = v_3(\sigma(n)).$$

For $0 \leq t \leq H$, put

$$M_t = \frac{\sigma(n)}{3^{H-t}}.$$

Since the primes p_i are odd, M_t is even. Lemma 5.1 therefore gives infinitely many primes $q_t > \sigma(n)$ satisfying its conclusion with $M = M_t$.

Let P_t denote the divisibility poset $S(nq_t)$. It consists of a lower copy x_I and an upper copy $(q_t + 1)x_J$ of B_r . There are no divisibilities from the upper copy back to the lower one because $q_t > \sigma(n)$. By (5.1), all primes other than 3 are automatically allowed by M_t , while

$$v_3\left(\frac{x_I}{\gcd(x_I, x_J)}\right) = \max\{W(I) - W(J), 0\}.$$

Hence the cross-relations are exactly

$$\boxed{x_I \mid (q_t + 1)x_J \iff W(I) - W(J) \leq t.} \quad (5.6)$$

Thus the arithmetic problem has been reduced to a one-parameter family of two-layer Boolean posets.

Lemma 5.3 (Width interpolation). *Let $\alpha_t = \text{width}(P_t)$. Then*

$$\alpha_{t+1} \leq \alpha_t \leq \alpha_{t+1} + 1 \quad (0 \leq t < H). \quad (5.7)$$

Moreover,

$$\alpha_0 = b_{r+1}, \quad \alpha_H = b_r. \quad (5.8)$$

Consequently every integer between b_r and b_{r+1} occurs among the values α_t .

Proof. For an integer s , let

$$I_s = \{I \subseteq [r] : W(I) \leq s\}, \quad F_s = \{I \subseteq [r] : W(I) > s\},$$

with the induced Boolean order. We claim that

$$\alpha_t = \max_{s \in \mathbb{Z}} (\text{width}(I_s) + \text{width}(F_{s+t})). \quad (5.9)$$

Indeed, split an antichain of P_t into its upper and lower parts. If the upper part is nonempty, choose an upper element x_J of largest weight and put $s = W(J)$. Every upper element then belongs to I_s . If a lower element x_I had $W(I) \leq s + t$, then (5.6) would give $x_I \mid (q_t + 1)x_J$, contradicting the antichain property. Hence every lower element belongs to F_{s+t} . The two parts therefore have sizes at most $\text{width}(I_s)$ and $\text{width}(F_{s+t})$. If the upper part is empty, take $s < -t$, so that I_s is empty and F_{s+t} is the whole Boolean lattice. If the lower part is empty, take $s \geq H$, so that I_s is the whole Boolean lattice and F_{s+t} is empty. Conversely, choose maximal antichains $A \subseteq I_s$ and $B \subseteq F_{s+t}$, place A in the upper copy and B in the

lower copy, and consider $I \in B$, $J \in A$. Then $W(I) > s + t \geq W(J) + t$, so (5.6) excludes every lower-to-upper comparability. Thus $A \cup B$ is an antichain. This proves (5.9).

The weights (5.5) are pairwise distinct integers. Therefore, for fixed s , passing from F_{s+t} to F_{s+t+1} deletes at most one element. Deleting one element from a finite poset can lower its width by at most one. Formula (5.9) then gives

$$\alpha_t - 1 \leq \alpha_{t+1} \leq \alpha_t,$$

which is (5.7).

For $t = H$, condition (5.6) holds for every pair I, J , so every lower element lies below every upper element. Thus P_H is the ordinal sum of two copies of B_r and

$$\alpha_H = b_r.$$

For $t = 0$, identify the lower copy with the layer of B_{r+1} having last coordinate 0, and the upper copy with the layer having last coordinate 1. Comparabilities within each layer are the Boolean ones. A cross-comparability in B_{r+1} has the form $(I, 0) \leq (J, 1)$ with $I \subseteq J$; then $W(I) \leq W(J)$, so (5.6) gives the same comparability in P_0 . Thus every comparability of B_{r+1} occurs in P_0 , and hence

$$\alpha_0 \leq b_{r+1}.$$

Put

$$h = \left\lfloor \frac{r+1}{2} \right\rfloor.$$

Every h -element subset has weight at least hA , whereas every $(h-1)$ -element subset has weight at most

$$(h-1)A + (A-1) = hA - 1.$$

Therefore the h -th rank of the lower copy together with the $(h-1)$ -st rank of the upper copy is an antichain in P_0 . Its size is

$$\binom{r}{h} + \binom{r}{h-1} = \binom{r+1}{h} = b_{r+1}.$$

Thus $\alpha_0 = b_{r+1}$. Since the integer sequence α_t decreases from b_{r+1} to b_r in steps of size at most one, it assumes every intermediate integer value. $\square$

Example 5.1 (The interpolation for $r = 3$). Here $b_3 = 3$ and $b_4 = 6$. The three weights in the construction are

$$9, 10, 12.$$

As the threshold t increases, the widths of the corresponding two-layer posets take the values

$$6, 5, 4, 3.$$

Thus the construction does not jump directly from the Boolean maximum 6 to 3; it passes through every intermediate width. Lemma 5.3 proves the same phenomenon for every r .

Theorem 5.4 (Complete squarefree spectrum). *For every $r \geq 1$,*

$$\mathcal{W}_r = \left\{ 1, 2, \dots, \binom{r}{\lfloor r/2 \rfloor} \right\}. \quad (5.10)$$

Moreover, every value in this interval is attained by infinitely many squarefree integers with exactly r prime factors.

Proof. We argue by induction on r . The assertion is immediate for $r = 1$. Assume it for r . Corollary 5.2 preserves every value $1, \dots, b_r$ when one new prime factor is added, with infinitely many realizations. Lemma 5.3 and the construction preceding it realize every value

$$b_r, b_r + 1, \dots, b_{r+1}$$

with $r + 1$ prime factors, again with infinitely many realizations because the final prime q_t can be chosen in infinitely many ways. Thus every value from 1 to b_{r+1} occurs. The upper bound in Theorem 3.1 shows that no larger value is possible. $\square$

Remark 5.5. The interpolation uses both arithmetic and combinatorial control. The private primes ℓ_i keep the original copy exactly Boolean, while the separated 3-adic valuations reduce every cross-divisibility to the single threshold $W(I) - W(J) \leq t$. Varying t then changes the width by at most one, which forces the spectrum to have no gaps.

6 An explicit cubefree family

The fixed-pattern theorem gives a particularly simple expression when all exponents are 1 or 2. If

$$n = p_1 \cdots p_s q_1^2 \cdots q_t^2,$$

then the relevant product is

$$D_2^s \times D_3^t.$$

The factors D_2 and D_3 have rank generating polynomials $1 + x$ and $1 + 2x$. They are normal with log-concave rank numbers, and these properties are preserved under direct products; see, for example, [8, Ch. 4]. Hence the width is the largest rank number.

Corollary 6.1. *For fixed $s, t \geq 0$,*

$$\max a(p_1 \cdots p_s q_1^2 \cdots q_t^2) = \max_k [x^k] (1 + x)^s (1 + 2x)^t.$$

In particular,

$$\max a(q_1^2 \cdots q_r^2) = \max_{0 \leq k \leq r} 2^k \binom{r}{k} \sim \frac{3^{r+1}}{2\sqrt{\pi r}}.$$

7 Optimal variance of the divisibility poset

Equal exponent patterns lead, through Theorem 4.1, to powers of D_m . To estimate their widths we use Engel's notion of a poset representation.

For a finite poset P with uniform weight, a *representation* is a function

$$x : P \rightarrow \mathbb{R}$$

such that

$$x(q) - x(p) \geq 1 \quad (p < q).$$

Let $V(P)$ be the minimum of the uniform variance $\text{Var}(x)$ over all representations x , and put

$$V_m = V(D_m).$$

Engel's asymptotic product theorem states that for every finite poset P which is not an antichain,

$$\text{width}(P^r) \sim \frac{|P|^r}{\sqrt{2\pi r V(P)}} \quad (r \rightarrow \infty)$$

under uniform weights [5, 8]; see in particular [8, Theorem 7.2.1]. Our task is therefore to determine $V(D_m)$.

7.1 A Ferrers lemma

For graded posets, the condition that every nonempty filter has average rank at least the global average is closely related to *rank compression*; see Bezrukavov and Engel [7]. Engel's optimal-representation theory also has linear-programming and flow formulations [5, 6]. Our Ferrers truncations need not be graded, so we use none of these results as a black box. We prove directly the filter-average inequality needed for optimality.

For integers

$$w_0 > w_1 > \cdots > w_{H-1} > 0, \quad w_H = 0,$$

let

$$P(w) = \{(a, b) : 0 \leq b < H, 0 \leq a < w_b\}$$

with the coordinatewise order, and define the natural height

$$\rho(a, b) = a + b.$$

Lemma 7.1 (Ferrers lemma). *Assume*

$$w_b - w_{b+1} \in \{1, 2\} \quad (0 \leq b < H).$$

Then ρ is an optimal representation of $P(w)$.

We prove the lemma by showing that every filter has average height at least the global mean. Then every monotone perturbation of ρ has nonnegative covariance with ρ , so the variance cannot decrease. The remaining work is geometric: row drops of 1 or 2 force the relevant Ferrers boundary pieces to have the required average height.

Proof. Let μ be the average of ρ on $P(w)$. We first prove that every order ideal has average height at most μ .

An ideal is described by row lengths

$$w_b \geq x_b \geq x_{b+1} \geq 0, \quad x_H := 0.$$

Relax $x_0, \dots, x_{H-1}$ to real variables and let K be the resulting compact polytope. Define

$$\Phi(x) = \sum_{b=0}^{H-1} \left(bx_b + \frac{x_b(x_b - 1)}{2} - \mu x_b \right).$$

For integral x , this is the total value of $\rho - \mu$ on the corresponding ideal. The function Φ is convex, so it is enough to prove $\Phi \leq 0$ at the vertices of K .

At a vertex, every maximal constant block

$$x_s = \cdots = x_t = c$$

is pinned either at $c = 0$ or at $c = w_t$. To see this carefully, suppose $0 < c < w_t$. Since the row bounds are decreasing, $c < w_t \leq w_b$ for every $s \leq b \leq t$, so none of the upper bounds inside the block is tight. Maximality of the constant block gives $x_{s-1} > c$ when $s > 0$ and $c > x_{t+1}$ when $t < H - 1$; at the boundary one instead has the strict inequality $c > 0$. Hence, for sufficiently small $\varepsilon > 0$, replacing every x_b in the block by $c + \varepsilon$ or by $c - \varepsilon$ remains in K . The point is therefore the midpoint of two distinct feasible points and cannot be a vertex.

A positive constant block $x_s = \dots = x_t = w_t$ contributes to the complement precisely the cells in rows $s, \dots, t - 1$ lying to the right of column $w_t - 1$; a final zero block contributes the remaining full rows. Thus the complement of a vertex ideal is a disjoint union of wedges

$$W_{s,t} = \{(a, b) : s \leq b < t, w_t \leq a < w_b\}, \quad 0 \leq s < t \leq H.$$

We show that every such wedge has average height at least μ .

Fix s . Inside $W_{s,t}$, the average height of row b is

$$b + \frac{w_t + w_b - 1}{2}, \quad s \leq b < t.$$

These row averages are nondecreasing in b , because $w_b - w_{b+1} \leq 2$. For $t < H$,

$$W_{s,t+1} = W_{s,t} \dot{\cup} R_{s,t},$$

where

$$R_{s,t} = \{(a, b) : w_{t+1} \leq a < w_t, s \leq b \leq t\}.$$

The smallest row average in $W_{s,t}$ exceeds the average height of $R_{s,t}$ by

$$\frac{s + w_s - w_{t+1} - t}{2} \geq \frac{1}{2},$$

because every row-length drop is at least 1, so $w_s - w_{t+1} \geq t - s + 1$. Hence adjoining $R_{s,t}$ can only decrease the average, and therefore

$$\text{av}(W_{s,t}) \geq \text{av}(W_{s,H}).$$

Now $W_{s,H}$ is simply the union of the full rows $s, \dots, H - 1$. The average height of the full b -th row is

$$b + \frac{w_b - 1}{2},$$

and these row averages are again nondecreasing. Removing the initial rows can therefore only increase the overall average, so

$$\text{av}(W_{s,H}) \geq \mu.$$

Thus every wedge has average height at least μ . The complement of every vertex ideal has average at least μ , hence every vertex ideal has average at most μ . Since Φ is convex, the same holds for every ideal.

Equivalently, every nonempty filter has average ρ at least μ . Let x be any representation and write

$$x = \rho + y.$$

The set $P(w)$ is an order ideal in $\mathbb{N}_0^2$, so every cover changes exactly one coordinate by 1. Hence ρ increases by exactly 1 along every cover relation, and y is nondecreasing. Every

nondecreasing function on a finite poset is a constant plus a nonnegative linear combination of indicator functions of filters: list its distinct values $c_0 < \cdots < c_m$ and write

$$y = c_0 + \sum_{j=1}^m (c_j - c_{j-1}) \mathbf{1}_{\{y \geq c_j\}},$$

where each upper level set $\{y \geq c_j\}$ is a filter. It follows from the filter inequality that

$$\text{Cov}(\rho, y) \geq 0.$$

Therefore

$$\text{Var}(x) = \text{Var}(\rho) + 2 \text{Cov}(\rho, y) + \text{Var}(y) \geq \text{Var}(\rho).$$

Thus ρ is optimal. □

7.2 The logarithmic Ferrers diagrams

For $T \geq 1$, define

$$Q_T = \{(a, b) \in \mathbb{N}_0^2 : 2^a 3^b \leq T\}.$$

Put

$$N(T) = |Q_T|, \quad \mu(T) = \frac{1}{N(T)} \sum_{(a,b) \in Q_T} (a + b),$$

and

$$v(T) = \frac{1}{N(T)} \sum_{(a,b) \in Q_T} (a + b - \mu(T))^2.$$

Figure 2 shows the shape to keep in mind.

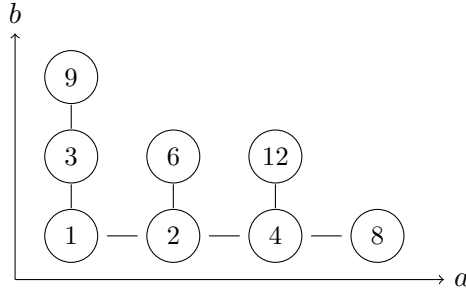


Figure 2: The diagram Q_{12} . The label at (a, b) is $2^a 3^b$. Moving right multiplies by 2, moving up multiplies by 3, and the natural height is $a + b$. The row lengths are 4, 3, 1.

The nonempty rows of Q_T have lengths

$$w_b = 1 + \left\lfloor \log_2 \frac{T}{3^b} \right\rfloor.$$

Since $1 < \log_2 3 < 2$, successive nonempty row lengths differ by 1 or 2; the last nonempty row has length 1 or 2. Lemma 7.1 therefore applies to every Q_T .

We need one quantitative consequence.

Lemma 7.2 (Scaling lemma). *For every integer $T \geq 1$,*

$$\mu(2T) - \mu(T) \geq \frac{1}{2}.$$

Consequently,

$$\mu(4T) - \mu(T) \geq 1.$$

Proof. Let h be the number of points of the first column of Q_{2T} , equivalently

$$h = \#\{b \geq 0 : 3^b \leq 2T\}.$$

For $0 \leq b < h$, let w_b be the number of $a \geq 0$ such that $2^a 3^b \leq T$; the last w_b is allowed to be 0. Put

$$N = N(T) = \sum_{b=0}^{h-1} w_b$$

and

$$c_b = w_b + 2b.$$

The sequence (w_b) is nonincreasing, while

$$c_{b+1} - c_b = 2 - (w_b - w_{b+1}) \in \{0, 1\}$$

whenever both sides are defined. Hence (c_b) is nondecreasing. Since (w_b) and (c_b) are oppositely ordered,

$$h \sum_b w_b c_b - N \sum_b c_b = \sum_{0 \leq i < j < h} (w_i - w_j)(c_i - c_j) \leq 0.$$

Therefore

$$\frac{\sum_b w_b c_b}{N} \leq \frac{1}{h} \sum_b c_b.$$

Now

$$2\mu(T) + 1 = \frac{\sum_b w_b c_b}{N},$$

whereas

$$\sum_b c_b = N + h(h-1).$$

Therefore

$$\mu(T) \leq \frac{1}{2} \left(\frac{N}{h} + h - 2 \right). \quad (7.1)$$

The set Q_{2T} is the disjoint union of the translate

$$(a, b) \mapsto (a+1, b)$$

of Q_T and the first column

$$\{(0, b) : 0 \leq b < h\}.$$

Thus

$$N(2T) = N + h$$

and

$$\mu(2T) - \mu(T) = \frac{N + h((h-1)/2 - \mu(T))}{N + h}.$$

By (7.1),

$$\frac{h-1}{2} - \mu(T) \geq \frac{h-N}{2h}.$$

The numerator above is therefore at least

$$N + \frac{h-N}{2} = \frac{N+h}{2},$$

which proves

$$\mu(2T) - \mu(T) \geq \frac{1}{2}.$$

Applying this twice gives the second assertion. $\square$

Example 7.1 (The scaling lemma at $T = 6$). The diagram Q_6 has row lengths 3, 2, so its rank values are

$$0, 1, 2; \quad 1, 2$$

and therefore $\mu(6) = 6/5$. The diagram Q_{12} has row lengths 4, 3, 1, with total rank sum 14, so

$$\mu(12) = \frac{14}{8} = \frac{7}{4}.$$

Thus

$$\mu(12) - \mu(6) = \frac{11}{20} > \frac{1}{2}.$$

The proof of Lemma 7.2 shows that the same half-unit gain holds uniformly for every T , even though the Ferrers shape changes whenever T crosses a number of the form $2^a 3^b$.

Lemma 7.3. *The function $T \mapsto \mu(T)$ is nondecreasing on the positive integers.*

Proof. If $S \leq T$, then Q_S is an order ideal of Q_T . By Lemma 7.1, every ideal of Q_T has average height at most the average height of Q_T . Hence $\mu(S) \leq \mu(T)$. $\square$

7.3 Only the primes 2 and 3 are active

At first sight there is no reason for the higher primes to be irrelevant. The full poset D_m contains divisibility constraints arising from multiplication by every prime $p \leq m$, and deleting all constraints with a factor $p \geq 5$ changes the order relation drastically. The next theorem shows that, for the optimal-variance problem, all of those deleted constraints are nevertheless redundant: the 2- and 3-directions already force the same minimum variance as the complete divisibility order.

Let $D_m^{(2,3)}$ be the poset on $\{1, \dots, m\}$ obtained by retaining only those divisibility relations whose quotient is of the form $2^c 3^d$. Every positive integer has a unique representation

$$n = u 2^a 3^b, \quad (u, 6) = 1.$$

For fixed u , the corresponding connected component of $D_m^{(2,3)}$ is isomorphic to

$$Q_{\lfloor m/u \rfloor}.$$

Theorem 7.4 (2, 3-reduction). *For every $m \geq 1$,*

$$\boxed{V(D_m) = V(D_m^{(2,3)})}.$$

More precisely, if

$$n = u2^a3^b, \quad (u, 6) = 1,$$

then

$$\boxed{x_m(n) = a + b - \mu\left(\left\lfloor \frac{m}{u} \right\rfloor\right)} \quad (7.2)$$

is an optimal representation of D_m .

Proof. By Lemma 7.1, $a+b$ is optimal on every component Q_T of $D_m^{(2,3)}$. Since the components are disconnected, they may be translated independently. Centering each component at mean 0 removes the between-component variance, so (7.2) is optimal for $D_m^{(2,3)}$.

It remains to prove that (7.2) satisfies every divisibility constraint of the full poset D_m . Suppose $n < n'$ and $n \mid n'$. Write

$$n = u2^a3^b, \quad n' = uv2^{a+c}3^{b+d},$$

where $(uv, 6) = 1$ and $c, d \geq 0$. Set

$$T = \left\lfloor \frac{m}{u} \right\rfloor, \quad T' = \left\lfloor \frac{m}{uv} \right\rfloor = \left\lfloor \frac{T}{v} \right\rfloor.$$

Then

$$x_m(n') - x_m(n) = c + d + \mu(T) - \mu(T').$$

If $c + d \geq 1$, Lemma 7.3 gives a difference at least 1.

It remains to consider $c = d = 0$. Then $v > 1$ and $(v, 6) = 1$, hence $v \geq 5$. Thus

$$T' \leq \left\lfloor \frac{T}{5} \right\rfloor, \quad 4T' \leq T.$$

Using Lemmas 7.3 and 7.2,

$$\mu(T) - \mu(T') \geq \mu(4T') - \mu(T') \geq 1.$$

Hence (7.2) is a representation of D_m .

Because D_m has all the constraints of $D_m^{(2,3)}$, one always has

$$V(D_m) \geq V(D_m^{(2,3)}).$$

Conversely, we have just exhibited an optimal representation of $D_m^{(2,3)}$ that is feasible for D_m , so

$$V(D_m) \leq V(D_m^{(2,3)}).$$

The two variances are equal, and (7.2) is optimal for the full divisibility poset. $\square$

Thus multiplication by primes $5, 7, 11, \dots$ creates no additional active constraint in the optimal variance problem. This is stronger than saying that the higher-prime constraints are asymptotically negligible: they are already redundant for the exact finite optimization problem for every m .

7.4 Why exactly 2 and 3?

The preceding theorem proves that the primes 2 and 3 suffice. We now show that this pair is sharp in two complementary senses.

For a set $\mathcal{P}$ of primes, let $D_m^{(\mathcal{P})}$ be the poset on $\{1, \dots, m\}$ in which $a \preceq b$ when $a \mid b$ and every prime divisor of b/a belongs to $\mathcal{P}$.

Proposition 7.5 (Minimal universal prime core). *For a set $\mathcal{P}$ of primes,*

$$\boxed{V(D_m^{(\mathcal{P})}) = V(D_m) \text{ for every } m \iff \{2, 3\} \subseteq \mathcal{P}.} \quad (7.3)$$

Proof. If $\{2, 3\} \subseteq \mathcal{P}$, then

$$D_m^{(2,3)} \subseteq D_m^{(\mathcal{P})} \subseteq D_m$$

in the sense of inclusion of order relations. Adding constraints can only increase the minimum representation variance, so Theorem 7.4 gives

$$V(D_m) = V(D_m^{(2,3)}) \leq V(D_m^{(\mathcal{P})}) \leq V(D_m).$$

Hence equality holds throughout.

Conversely, if $2 \notin \mathcal{P}$, then $D_2^{(\mathcal{P})}$ is an antichain and has optimal variance 0, whereas D_2 is a two-element chain and has optimal variance $1/4$. Thus universal equality fails. Suppose next that $2 \in \mathcal{P}$ but $3 \notin \mathcal{P}$. Then $D_3^{(\mathcal{P})}$ consists of the chain $1 < 2$ and the isolated point 3, so its optimal variance is

$$\frac{(-1/2)^2 + (1/2)^2 + 0^2}{3} = \frac{1}{6}.$$

For the full poset D_3 , the constraints are $1 < 2$ and $1 < 3$; the centered optimal representation has values $-2/3, 1/3, 1/3$, and hence

$$V(D_3) = \frac{2}{9} > \frac{1}{6}.$$

Thus 3 is also necessary. □

The same phenomenon has a local two-prime form. Let $p < q$ be distinct primes and define

$$Q_T(p, q) = \{(a, b) \in \mathbb{N}_0^2 : p^a q^b \leq T\}$$

with the coordinatewise order and natural height

$$\rho(a, b) = a + b.$$

Theorem 7.6 (Sharp Ferrers-slope threshold). *For a real number $\theta > 1$ and $c \geq 0$, define*

$$Q_c(\theta) = \{(a, b) \in \mathbb{N}_0^2 : a + \theta b \leq c\}$$

with the coordinatewise order and natural height $\rho(a, b) = a + b$. Then ρ is an optimal representation of $Q_c(\theta)$ for every $c \geq 0$ if and only if

$$\boxed{1 < \theta \leq 2.} \quad (7.4)$$

Proof. Assume first that $1 < \theta \leq 2$. The nonempty rows of $Q_c(\theta)$ have lengths

$$w_b = 1 + \lfloor c - \theta b \rfloor.$$

Consecutive nonempty row lengths differ by 1 or 2, and the last nonempty row has length 1 or 2. Lemma 7.1 therefore applies for every c .

Conversely, suppose $\theta > 2$ and put

$$c = \max\{\theta, 3\}.$$

Then $c < \theta + 1$ and $c < 2\theta$. Hence $Q_c(\theta)$ consists of the horizontal chain

$$(0, 0), (1, 0), \dots, (k, 0), \quad k = \lfloor c \rfloor \geq 3,$$

together with the single maximal point $(0, 1)$. The filter $\{(0, 1)\}$ has average height 1, while the global average is

$$\frac{0 + 1 + \dots + k + 1}{k + 2} > 1.$$

If ρ were optimal, every filter would have average height at least the global average: otherwise adding a sufficiently small positive multiple of the filter indicator would preserve all representation inequalities and decrease the variance to first order. Thus ρ is not optimal. $\square$

Remark 7.7. The threshold 2 has a direct geometric meaning. It is the largest slope for which every successive row-length drop in the Ferrers diagrams $Q_c(\theta)$ is at most 2; once $\theta > 2$, a drop of at least 3 can occur, and the natural height need no longer be optimal.

The two-prime statement is now an immediate logarithmic specialization.

Corollary 7.8 (Sharp two-prime Ferrers threshold). *For distinct primes $p < q$, the natural height is an optimal representation of*

$$Q_T(p, q) = \{(a, b) \in \mathbb{N}_0^2 : p^a q^b \leq T\}$$

for every $T \geq 1$ if and only if

$$\boxed{q < p^2}. \tag{7.5}$$

Proof. After dividing $\log T$ by $\log p$, the condition $p^a q^b \leq T$ becomes

$$a + (\log_p q)b \leq \log_p T.$$

Apply Theorem 7.6. Since $p < q$, one has $\log_p q > 1$, and optimality for every truncation is equivalent to $\log_p q \leq 2$, that is $q \leq p^2$. Equality is impossible for distinct primes, so this is exactly $q < p^2$. $\square$

There is also a sharp scale interpretation. Write $\mu_{p,q}(T)$ for the average of $a + b$ on $Q_T(p, q)$.

Corollary 7.9 (Critical dilation). *If $p < q < p^2$, then*

$$\mu_{p,q}(pT) - \mu_{p,q}(T) \geq \frac{1}{2} \quad (T \geq 1), \tag{7.6}$$

and the least positive integer L such that

$$\mu_{p,q}(LT) - \mu_{p,q}(T) \geq 1 \quad \text{for every } T \geq 1$$

is

$$\boxed{L = p^2}. \tag{7.7}$$

Proof. The proof of Lemma 7.2 uses only that the Ferrers row lengths drop by 1 or 2 and that multiplication of T by the first prime translates one copy of the diagram by one step in the a -direction while adding a first column. The same argument therefore gives (7.6). Iterating twice yields

$$\mu_{p,q}(p^2T) - \mu_{p,q}(T) \geq 1.$$

For the reverse inequality, take $T = 1$ and $L < p^2$. Since p^2 , pq , and q^2 all exceed L , the set $Q_L(p, q)$ contains at most the three points corresponding to $1, p, q$, of ranks $0, 1, 1$. Hence

$$\mu_{p,q}(L) < 1 = \mu_{p,q}(1) + 1.$$

Thus no smaller integer dilation works uniformly. $\square$

For $(p, q) = (2, 3)$ the critical dilation is therefore 4. This also explains the reduction theorem: the prime 3 still lies below the unit-gain scale 4, whereas the next prime is already $5 > 4$. Moving between distinct $(2, 3)$ -components therefore costs enough centering distance to absorb every prime factor ≥ 5 .

A second characterization shows that adding any further prime destroys the uniform optimality of the natural exponent-sum height itself. For a set $\mathcal{P}$ of primes and $T \geq 1$, let

$$S_{\mathcal{P}}(T) = \{n \leq T : \text{every prime divisor of } n \text{ belongs to } \mathcal{P}\},$$

ordered by divisibility, and put $\rho(n) = \Omega(n)$.

Corollary 7.10 (A maximal natural-height core). *Let $2 \in \mathcal{P}$. Then $\rho(n) = \Omega(n)$ is an optimal representation of $S_{\mathcal{P}}(T)$ for every T if and only if*

$$\boxed{\mathcal{P} \subseteq \{2, 3\}}. \tag{7.8}$$

Proof. For $\mathcal{P} = \{2\}$ the truncations are chains, and for $\mathcal{P} = \{2, 3\}$ the assertion is Theorem 7.8. These also cover every set $\mathcal{P}$ with $2 \in \mathcal{P} \subseteq \{2, 3\}$.

Now suppose that $\mathcal{P}$ contains a prime $r \geq 5$ and set

$$T = \max\{8, r\} < 2r.$$

Then r is a maximal element of $S_{\mathcal{P}}(T)$, so $\{r\}$ is a filter of average height 1. On the other hand the five points

$$1, 2, 4, 8, r$$

have heights $0, 1, 2, 3, 1$. Every other point has height at least 1, and therefore the global average height is strictly larger than 1. As in the proof of Theorem 7.8, the filter criterion rules out optimality of ρ . $\square$

Together, the two corollaries characterize $\{2, 3\}$ from both sides. It is the smallest universal set of prime directions that determines the exact variance of every D_m , and the largest prime set containing 2 for which the natural exponent-sum height is optimal on every truncation.

7.5 Exact formula and limiting constant

Theorem 7.4 immediately gives an exact finite formula.

Corollary 7.11. *For every $m \geq 1$,*

$$V_m = \frac{1}{m} \sum_{\substack{u \leq m \\ (u,6)=1}} N\left(\left\lfloor \frac{m}{u} \right\rfloor\right) v\left(\left\lfloor \frac{m}{u} \right\rfloor\right). \quad (7.9)$$

Proof. Each component indexed by u has $N(\lfloor m/u \rfloor)$ elements and, after centering, variance $v(\lfloor m/u \rfloor)$. All component means are 0, so the variance of the union is the size-weighted average of the component variances. $\square$

Theorem 7.12. *The sequence V_m converges. More precisely,*

$$V_m = C + O\left(\frac{(\log(2m))^4}{\sqrt{m}}\right),$$

where

$$C = \frac{1}{3} \sum_{t \geq 1} \frac{N(t)v(t)}{t(t+1)}. \quad (7.10)$$

Numerically,

$$C = 1.036793420159825251 \dots$$

Proof. Put

$$F(t) = N(t)v(t).$$

Since both coordinates in Q_t are $O(\log(2t))$,

$$N(t) = O((\log(2t))^2), \quad v(t) = O((\log(2t))^2),$$

and therefore

$$F(t) = O((\log(2t))^4).$$

The series in (7.10) is consequently absolutely convergent.

Group (7.9) according to

$$t = \left\lfloor \frac{m}{u} \right\rfloor.$$

Let

$$A_m(t) = \#\left\{u : (u,6) = 1, \frac{m}{t+1} < u \leq \frac{m}{t}\right\}.$$

By periodicity modulo 6,

$$\#\{u \leq x : (u,6) = 1\} = \frac{x}{3} + O(1)$$

with an absolute error bound uniform in x . Consequently, uniformly in m and t ,

$$A_m(t) = \frac{m}{3t(t+1)} + O(1).$$

Hence

$$V_m = \sum_{t \geq 1} F(t) \frac{A_m(t)}{m}.$$

For $t \leq \sqrt{m}$, replacing $A_m(t)/m$ by $1/(3t(t+1))$ contributes

$$O\left(\frac{1}{m} \sum_{t \leq \sqrt{m}} (\log(2t))^4\right) = O\left(\frac{(\log(2m))^4}{\sqrt{m}}\right).$$

For $t > \sqrt{m}$, the original sum corresponds to $u < \sqrt{m}$, so there are only $O(\sqrt{m})$ terms and its total contribution is

$$O\left(\frac{(\log(2m))^4}{\sqrt{m}}\right).$$

Finally,

$$\sum_{t > \sqrt{m}} \frac{F(t)}{t(t+1)} = O\left(\frac{(\log(2m))^4}{\sqrt{m}}\right).$$

Combining these estimates proves the result. For the displayed decimal, the series can be grouped between consecutive integers of the form $2^a 3^b$: on each such interval $N(t)$ and $v(t)$ are constant, while

$$\sum_{t=A}^B \frac{1}{t(t+1)} = \frac{1}{A} - \frac{1}{B+1}.$$

The remaining tail is bounded by the estimate $F(t) = O((\log(2t))^4)$ used above. $\square$

Table 1 illustrates the convergence. The values are obtained directly from the exact formula (7.9), not from an asymptotic approximation.

m	V_m	m	V_m
10	0.6214285714	2000	1.0231569379
20	0.7700000000	5000	1.0316806397
50	0.8716190476	10000	1.0339721278
100	0.9379761905	20000	1.0352570801
200	0.9789142857	50000	1.0360588433
500	0.9941762923	100000	1.0363963936
1000	1.0127665761	200000	1.0365806287
500000	1.0366853936	1000000	1.0367353914
∞	1.036793420159825251...		

Table 1: Selected values of the optimal variance V_m , computed from the exact finite formula (7.9).

Already at $m = 10^6$, the value in Table 1 differs from C by about 5.8×10^{-5} .

Remark 7.13. There is also a probabilistic interpretation of the constant. The weights

$$p_t = \frac{N(t)}{3t(t+1)}$$

form a probability distribution, because

$$\sum_{t \geq 1} \frac{N(t)}{t(t+1)} = \sum_{a,b \geq 0} \frac{1}{2^a 3^b} = 3.$$

Choose T with distribution (p_t) and then choose (A, B) uniformly from Q_T . Then

$$\Pr(A = a, B = b) = \frac{1}{3} \frac{1}{2^a 3^b} = \frac{1}{2^{a+1}} \frac{2}{3^{b+1}},$$

so A and B are independent geometric random variables. In particular,

$$\text{Var}(A + B) = 2 + \frac{3}{4} = \frac{11}{4},$$

while

$$C = \mathbb{E}[\text{Var}(A + B \mid T)].$$

The law of total variance therefore gives the more intuitive identity

$$\frac{11}{4} = C + \text{Var}(\mu(T)).$$

Thus C is the part of the product-geometric variance that remains after each logarithmic Ferrers component has been centered. Numerically, the removed between-component variance is

$$\text{Var}(\mu(T)) = 1.713206579840174748 \dots$$

This also explains why the limiting constant is much smaller than $11/4$: the optimal representation is allowed to translate the different $(2, 3)$ -components independently before the full divisibility constraints are restored.

7.6 Equal exponent patterns

For fixed $e \geq 1$, define

$$M_e(r) = \max_{p_1, \dots, p_r} a(p_1^e \cdots p_r^e).$$

By Theorem 4.1,

$$M_e(r) = \text{width}(D_{e+1}^r).$$

Engel's asymptotic product theorem therefore yields the following.

Corollary 7.14. *For every fixed $e \geq 1$,*

$$M_e(r) \sim \frac{(e+1)^r}{\sqrt{2\pi r V_{e+1}}} \quad (r \rightarrow \infty).$$

Moreover,

$$\lim_{e \rightarrow \infty} \lim_{r \rightarrow \infty} \sqrt{r} \frac{M_e(r)}{(e+1)^r} = \frac{1}{\sqrt{2\pi C}} = 0.391799565148234 \dots$$

8 Minimum widths

The maximal problem is now completely combinatorial: Theorem 4.1 reduces it to P_e . The minimum behaves differently, because extra cross-divisibilities now help rather than hurt. This is where the arithmetic of repunits in different bases becomes unavoidable.

Define

$$L(e_1, \dots, e_r) = \min_{p_1, \dots, p_r} a(p_1^{e_1} \cdots p_r^{e_r}),$$

where the minimum is over distinct primes.

Proposition 8.1. *For every exponent vector $\mathbf{e}$,*

$$\boxed{L(\mathbf{e}, 1) = L(\mathbf{e}).}$$

Proof. The inequality $L(\mathbf{e}, 1) \geq L(\mathbf{e})$ follows from Proposition 2.1. For the reverse inequality, choose

$$m = p_1^{e_1} \cdots p_r^{e_r}$$

with $a(m) = L(\mathbf{e})$ and put

$$M = \text{lcm}\{\sigma(d) : d \mid m\}.$$

Choose a prime $q \nmid m$ satisfying

$$q \equiv -1 \pmod{M}.$$

Then

$$S(mq) = S(m) \cup (q+1)S(m),$$

and every element of the first copy divides every element of the second. A chain partition of $S(m)$ into $L(\mathbf{e})$ chains therefore extends to the union without increasing the number of chains. Hence $a(mq) \leq L(\mathbf{e})$, proving equality. $\square$

Proposition 8.2. *For every exponent vector $\mathbf{e}$,*

$$\boxed{L(\mathbf{e}, 2) \leq 2L(\mathbf{e}).}$$

Proof. Use the same m, M, q as above. Then

$$S(mq^2) = S(m) \cup (q+1)S(m) \cup (q^2+q+1)S(m).$$

The first two copies are covered by $L(\mathbf{e})$ chains, while the third copy is covered by another $L(\mathbf{e})$ chains. Thus $a(mq^2) \leq 2L(\mathbf{e})$. $\square$

In particular,

$$L(e, 2) \leq 2 \left\lceil \frac{e+1}{2} \right\rceil.$$

9 Computation and the remaining problem

The computations used in this paper are exact. To evaluate $a(n)$, form the distinct set $S(n)$, build the bipartite graph of its strict divisibility relation, and apply Dilworth's theorem. If $v = |S(n)|$ and a maximum matching has size m , then

$$a(n) = v - m.$$

No heuristic search for a large antichain is involved.

For squarefree integers with r prime factors, the values in $S(n)$ are subset products of the r numbers $p_i + 1$. Hence at most 2^r values enter the matching computation before possible collisions are removed. Before Theorem 5.4 was found, explicit searches produced every predicted width through $r = 7$. After the proof was obtained, we used these data as an independent check: exact matching reproduces the complete intervals 1–1, 1–2, 1–3, 1–6, 1–10, 1–20, and 1–35 for $r = 1, \dots, 7$. These computations are not used in the proof of the theorem.

The minimum problem already involves divisibility relations between repunits in different prime bases. Direct searches in the first cases agree with the upper bound of Proposition 8.2, but the cross-divisibilities make a general lower bound substantially harder. We do not attach a numerical verification range to this conjecture here, because the available calculations were exploratory rather than a systematic exhaustive test. We therefore record the natural equality as a conjecture rather than pursuing prime-by-prime case analysis.

Conjecture 9.1. *For every $e \geq 1$,*

$$L(e, 2) = 2 \left\lceil \frac{e+1}{2} \right\rceil.$$

More generally, the problem is to determine or estimate $L(e_1, \dots, e_r)$. The contrast with Theorem 4.1 is sharp: the maximum depends only on the universal index poset P_e , whereas the minimum measures how strongly cross-divisibilities between different repunit families can collapse that poset. Thus the maximal problem and the squarefree spectrum are now complete; the remaining arithmetic question is how far cross-divisibilities can collapse the universal product posets.

References

- [1] P. Erdős, *Note on sequences of integers no one of which is divisible by any other*, J. London Math. Soc. **10** (1935), 126–128.
- [2] J. D. Lichtman, *A proof of the Erdős primitive set conjecture*, Forum Math. Pi **11** (2023), e18. <https://doi.org/10.1017/fmp.2023.16>.
- [3] R. P. Dilworth, *A decomposition theorem for partially ordered sets*, Ann. of Math. (2) **51** (1950), 161–166.
- [4] P. G. L. Dirichlet, *Beweis des Satzes, dass jede unbegrenzte arithmetische Progression, deren erstes Glied und Differenz ganze Zahlen ohne gemeinschaftlichen Factor sind, unendlich viele Primzahlen enthält*, Abh. Königl. Preuss. Akad. Wiss. (1837), 45–71.
- [5] K. Engel, *Optimal representations of partially ordered sets and a limit Sperner theorem*, European J. Combin. **7** (1986), 287–302. [https://doi.org/10.1016/S0195-6698\(86\)80003-8](https://doi.org/10.1016/S0195-6698(86)80003-8).
- [6] K. Engel, *An algorithm for the determination of the variance of a partially ordered set*, J. Algorithms **19** (1995), 441–448. <https://doi.org/10.1006/jagm.1995.1046>.
- [7] S. L. Bezrukov and K. Engel, *Properties of graded posets preserved by some operations*, in R. L. Graham and J. Nešetřil (eds.), *The Mathematics of Paul Erdős II*, Algorithms and Combinatorics 14, Springer, Berlin, Heidelberg, 1997, 79–85. https://doi.org/10.1007/978-3-642-60406-5_8.
- [8] K. Engel, *Sperner Theory*, Encyclopedia of Mathematics and its Applications 65, Cambridge University Press, 1997.
- [9] D. Lewis and V. Souza, *The order dimension of divisibility*, J. Combin. Theory Ser. A **179** (2021), 105391. <https://doi.org/10.1016/j.jcta.2020.105391>.
- [10] M. Haiman, *The dimension of divisibility orders and multiset posets*, Order **41** (2024), 693–707. <https://doi.org/10.1007/s11083-023-09653-7>.
- [11] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A000203, A173439, A184395, A206028, A206030, <https://oeis.org>.
- [12] E. Sperner, *Ein Satz über Untermengen einer endlichen Menge*, Math. Z. **27** (1928), 544–548.